

CRYPTEX: fine-grained CRYPTocurrency datasets EXploration

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Abstract

When simulating financial models, fine-grained datasets significantly impact model effectiveness. Our research aimed to experiment with cryptocurrency time-series data, particularly candlestick data from exchanges like Binance. However, obtaining high-quality, one-second granularity data from exchange APIs and even online websites proved difficult, un-intuitive, impossible, or expensive. This challenge inspired us to create an efficient Python-based framework that extracts the transaction history for 153 cryptocurrency trading pairs from Binance.us since September 2019. This data is then cleaned and summarized into a variety of sub-datasets ranging from yearly to one-second granularity candlesticks. To enable ease of data-sharing, we published a sample (4-year BTCUSDT trading pair data) dataset on Kaggle and the entirety of the datasets in CSV formatted files (261GB) on a publicly accessible web server that updates nightly.

1 Introduction

Bitcoin [2] is a decentralized digital currency and the first-ever cryptocurrency created in 2009 by an unknown person or group using the pseudonym Satoshi Nakamoto. It operates on a decentralized peer-to-peer network, known as the blockchain, without the need for a central authority or intermediary. Bitcoin represents a transformative innovation in the financial landscape, and its impact has extended beyond the world of finance, inspiring the development of thousands of other cryptocurrencies and driving discussions about the future of money and decentralized technologies.

Like Bitcoin, there are thousands of cryptocurrencies that have been developed with different approaches and different use-cases. Analyzing cryptocurrency pricing data is going to be of high interest to the Fintech community. Getting access to this price data is readily available through both graphs and numerical datasets (e.g. CSV format). The sources that are available make it incredibly difficult to get OHLC (also known as candle stick data [1]) and volume data at fine-grained resolution below 1-min intervals.

2 Making a Case for Fine-Grained Data

Time-series data represent events or measurements, reflecting cryptocurrency prices over time. Candlestick plots visualize this data concisely, displaying essential values for a specific time period. Smaller candlestick data granularity means more opportunities to buy and sell, leading to greater potential profit.

For example, in Table 1 we explored the BTCUSDT candle stick data from 2021; if one purchased one bitcoin on 1/1/21, the potential profit (assuming an all knowing oracle that could identify the high and low of every candle stick) showed the highest profit with the 1-second granularity (nearly 400X higher than when using weekly candle stick data).

After a thorough review of the potential sources of data for 1-sec granularity candle sticks, we concluded that there were no sources that offered the 1-sec candle stick datasets for free. The only resource we found was from Binance.com through a web GUI interface. Downloading the BTCUSDT dataset took multiple tries, yielded 48 individual files (1-month period each), and did not separate the US-based transactions from the global transactions found at Binance.com.

Table 1. Financial Performance for each Granularity

Period	Potential Profit	Opportunities
1-sec	\$170M	28.9M
1-min	\$37M	524K
1-hour	\$5.4M	8.7K
1-day	\$1.1M	365
1-week	\$431K	52

3 Proposed Solution

This project aimed to download transactions (from September 2019 to July 2023) for all 153 trading pairs (see Figure 1) from Binance.us, and then generate candle stick data at various intervals such as 1-sec, 1-min, 1-hour, 1-day, 1-week, 1-month, and 1-year. All transactions and candle stick data is updated nightly for all trading pairs, and made available on a publicly available web server [5] hosted on the Mystic system[4]. We also published the Bitcoin dataset to Kaggle under "Fine-Grained Bitcoin Historical Dataset 2019-2023"[6].

1INCH, AAVE, ACH, ADA, ALGO, ALICE, ALPINE, ANKR, ANT, APE, API3, APT, ARB, ASTR, ATOM, AUDIO, AVAX, AXL, AXS, BAL, BAND, BAT, BCH, BICO, BLUR, BNB, BNT, BOND, BOSON, BTC, BTRST, CELO, CELR, CHZ, CLV, COMP, COTI, CRV, CTSI, DAI, DAR, DASH, DGB, DIA, DOGE, DOT, EGLD, ENJ, ENS, EOS, ETC, ETH FET, FIL, FLOKI, FLOW, FLUX, FORTH, FTM, GAL, GALA, GLM, GRT, GTC, HBAR, ICP, ICX, ILV, IMX, IOST, IOTA, JAM, KAVA, KDA, KNC, KSM, LAZI, LDO, LINK, LOKA, LOOM, LPT, LRC, LSK, LTC, LTO, MANA, MASK, MATIC, MKR, MXC, NEAR, NEO, NMR, OCEAN, OGN, OMG, ONE, ONT, OP, OXT, PAXG, POLYX, POND, PORTO, PROM, QNT, QTUM, RAD, RARE, REEF, REN, REQ, RLC, RNDR, ROSE, RVN, SAND, SANTOS, SHIB, SKL, SLP, SNX, SOL, STG, STORJ, SUSHI, SYS, T, THETA, TLM, TRAC, TUSD, UNI, USDC, VET, VITE, VOXEL, VTHO, WAVES, WAXP, XEC, XLM, XNO, XRP, XTZ, YFI, ZEC, ZEN, ZIL, ZRX

Figure 1. 153 trading pairs available for download [5]

Our solution involves a 5 stage pipeline from historical transaction extraction, summarizing the data in candlestick comma delimited text format, visualization of the data through interactive plots using plotly library, automation of the extraction process to handle failures, throttling due to limitations of the Binance.us API, incremental daily updates of the datasets, and finally the hosting of the datasets on Kaggle and our own web server.

Throughout the project, we encountered multiple challenges. Jupyter Notebook had limitations with large files and rendering large and complex interactive plots, which prompted us to switch to a command line interface. Additionally, we faced issues with Binance.com, due to access restrictions from a US-based IP address; therefore, we used a REST API to Binance.us and focused on US-based trading pair transactions. To address time zone issues, we used the Python's lambda library.

4 Evaluation

We initially considered GitHub to host our website and datasets; we found storage limitations (5 GB) insufficient for our data that currently is 261GB large. The framework we built runs continuously and gets daily updates to the datasets, including redrawing the plots. We run the website on the Mystic [4] Cloud at IIT that has 10GbE network connectivity and a 1.2TB RAID5 array spanning 6x SSD storage devices. To share our datasets, we made a website for all our data, utilizing HTML and CSS. We also established a page on Kaggle, where data scientists can access a data sample from Bitcoin since 09/2019 (see Figure 2).

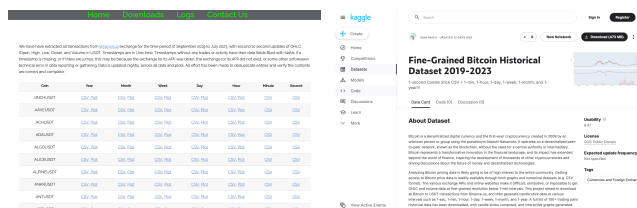


Figure 2. The Website and Kaggle page

We evaluated the total time needed to download 4-years of data on the BTCUSDT trading pair, using different approaches (see Table 2). The manual process from Binance.com took us about 1 hour and several attempts, including the manual merging of 48 files to assemble the final BTCUSDT candle stick data at different granularities. The Binance.us API is the approach we proposed, which involved the implementation of a Python-based framework that extracted all transactions, summarized the data with candle sticks, and generated interactive plots. Our approach took 660 seconds to download the entire 4-year dataset, including the summarizing of the data, and plot generation. Once we have the datasets published, others can get the same dataset using simple tools such as wget in a mere 51 seconds. We are experimenting with a time-series database XStore [3], that allows the efficient data extraction of subsets based on time index if the entire dataset is not required.

Table 2. Comparison of Data Collection Approaches

Source	Approach	# of files	Time (s)	Info
Binance.com	Manual	48	3,600	User-friendly, long run time
Binance.us	Manual	N/A	N/A	N/A
Binance.com	API	N/A	N/A	Not possible without VPN
Binance.us	API	1	660	Automated, requires programming
crypto.cs.iit.edu	XStore	1	70	Low latency, slower than wget
crypto.cs.iit.edu	wget	1	51	Fastest, no date range

5 Conclusions and Future Work

We have developed candlestick datasets at different granularity (1-sec, 1-min, 1-hour, 1-day, 1-week, 1-month, and 1-year) that are updated daily. These datasets encompass 153 trading pairs, spanning from September 2019 to the present day, and are now accessible to the public through our dedicated Kaggle page and website hosted on Mystic [4]. In the future, we intend to collect global transaction data from Binance.com and utilize these datasets for backtesting trading strategies. Furthermore, we aim to explore data correlation between trading pairs to improve backtesting accuracy. Furthermore, we are considering the implementation of sentiment analysis to predict a coin's price in the future.

References

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